Chapter 3: Crosstalk

1 Crosstalk

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1.1 3.1 Introduction *(p.3)*

Crosstalk is the unwanted coupling of energy from one signal line (the **aggressor/driver**) onto an adjacent line (the **victim**).

Two effects: 1. Changes impedance and propagation velocity affects system timing and signal integrity. 2. Induces noise voltage/current on the victim line degrades noise margins.

Depends on: data patterns, line-to-line spacing, and switching rates.

1.2 3.2 Mutual Inductance *(p.4)*

Magnetic-field coupling between lines is modelled by **mutual inductance** L_m.

When the aggressor carries a changing current, a voltage noise is induced on the victim:

$$V_{\text{noise},L_m} = L_m \frac{dI_{\text{driver}}}{dt}$$

(chap3.pdf, p.4)

1.3 3.3 Mutual Capacitance *(p.5)*

Electric-field coupling between lines is modelled by **mutual capacitance** C_m.

When the aggressor has a changing voltage, a current noise is injected onto the victim:

$$I_{\text{noise},C_m} = C_m \frac{dV_{\text{driver}}}{dt}$$

(chap3.pdf, p.5)

1.4 3.4 Inductance and Capacitance Matrices *(pp.68)*

For an N-conductor system, the parasitics are described by matrices.

1.4.1 Inductance matrix:

$$[L] = \begin{bmatrix} L_{11} & L_{12} & \cdots \\ L_{21} & L_{22} & \cdots \\ \vdots & & \ddots \end{bmatrix}$$

- L_{NN}: self-inductance of line N
- L_{MN}: mutual inductance between lines M and N

(chap3.pdf, p.6)

1.4.2 Capacitance matrix:

$$[C] = \begin{bmatrix} C_{11} & C_{12} & \cdots \\ C_{21} & C_{22} & \cdots \\ \vdots & & \ddots \end{bmatrix}$$

- C_{NN}: total capacitance seen by line N (capacitance to ground + all mutual capacitances)
- $C_{NM} = -C_m$ (mutual capacitance; negative off-diagonal)

(chap3.pdf, p.7)

1.4.3 Two-conductor example *(p.8)*:

For two lines with mutual capacitance C_m between them and ground capacitances:

$$C_{11} = C_{22} = C_{\text{self}} + C_m, \quad C_{12} = C_{21} = -C_m$$

C_m is obtained by grounding line 2 and measuring charge on line 1.

1.5 3.5 Field Simulators *(pp.910)*

Type	Speed	Accuracy	Use
Electrostatic (2D)	Fast (seconds)	Limited to simple geometries, no skin effect	Early design
Full-wave (3D)	Slow (hoursdays)	Handles complex geometry, frequency-dependent losses, dispersion, radiation	Final verification

Full-wave output is typically in **S-parameters** less directly useful for digital interconnect simulation.

1.6 3.6 Near-End and Far-End Crosstalk *(pp.1112)*

Near-End Crosstalk (NEXT / Backward Crosstalk): measured at the **driven end** of the victim line. Current is the **sum** of inductance- and capacitance-induced components:

$$I_{\text{near}} = I(L_m) + I_{\text{near}}(C_m)$$

(chap3.pdf, p.11)

Far-End Crosstalk (FEXT / Forward Crosstalk): measured at the **far end** of the victim line. Current is the **difference**:

$$I_{\text{far}} = I_{\text{far}}(C_m) - I(L_m)$$

(chap3.pdf, p.11)

Timing: - NEXT begins at $t = 0$, duration $= 2\,\text{TD}$ - FEXT occurs at $t = \text{TD}$, duration $\approx \tau_r$ (rise/fall time)

(chap3.pdf, p.12)

1.7 3.7 Crosstalk Magnitudes Matched Terminations *(pp.1314)*

For matched terminations ($R = Z_0$ at both ends), $\text{TD} = X\sqrt{LC}$, where X = line length, L, C = self-inductance and capacitance per unit length.

1.7.1 Case: $T_r < 2\text{TD}$ (long-line case)

NEXT peak amplitude:

$$A_{\text{NEXT}} = \frac{V_{\text{input}}}{4}\left(\frac{L_m}{L} + \frac{C_m}{C}\right)$$

(chap3.pdf, p.13)

FEXT peak amplitude:

$$B_{\text{FEXT}} = -\frac{V_{\text{input}} \cdot X\sqrt{LC}}{2T_r}\left(\frac{L_m}{L} - \frac{C_m}{C}\right)$$

(chap3.pdf, p.13)

Key insight: In the long-line case, NEXT is **independent of line length**, while FEXT **always depends on rise time and length**.

1.7.2 Case: $T_r > 2$TD (short-line case)

NEXT fails to reach maximum amplitude multiply NEXT by 2TD/T_r.
FEXT equations remain unchanged. *(p.14)*

1.8 3.8 Mismatched Terminations *(p.15)*

When the victim line termination $R \neq Z_0$, reflections add to the crosstalk:

$$V_x = V_{\text{crosstalk}} \cdot \frac{1+\rho}{1}$$

where $\rho = (R - Z_0)/(R + Z_0)$ and V_x is the adjusted near- or far-end noise. *(p.15)*

1.9 3.9 Example 3.2 Matched Terminations *(pp.1718)*

$Z_0 \approx 70\,\Omega$, $R = 70\,\Omega$, $V_{\text{in}} = 1.0$ V, $T_r = 100$ ps, $X = 2$ in.

From L and C matrices given:

$$A_{\text{NEXT}} = \frac{V_{\text{in}}}{4}\left(\frac{L_m}{L} + \frac{C_m}{C}\right), \quad B_{\text{FEXT}} = -\frac{V_{\text{in}} X \sqrt{LC}}{2T_r}\left(\frac{L_m}{L} - \frac{C_m}{C}\right)$$

(chap3.pdf, pp.1718)

1.10 3.10 SPICE Equivalent Circuit Model for Coupled Lines *(pp.2023)*

For a pair of coupled lines, the **mutual inductance** is modelled by a coupling factor K:

$$K = \frac{L_{12}}{\sqrt{L_{11}L_{22}}}$$

(chap3.pdf, p.21)

1.10.1 Example 3.4 Coupled Line Model *(pp.2123)*

Given: 5 in. long coupled lines, $T_r = 100$ ps. Key results: - Minimum segments: $\geq 10(x/v)/T_r$ 67 segments - Per-segment values: $L(N)$, $C_{1g}(N) = C_{11} - C_m$, $C_m(N)$ computed by scaling by 5/67 in./segment

(chap3.pdf, pp.2223)

1.11 3.11 Odd Mode Propagation *(pp.2427)*

When two adjacent lines carry **opposite-polarity signals** ($I_2 = -I_1$, $V_2 = -V_1$):

- **Odd-mode inductance** (reduced):

$$L_{\text{odd}} = L_{11} - L_m = L_{11} - L_{12}$$

(chap3.pdf, p.25)

- **Odd-mode capacitance** (increased):

$$C_{\text{odd}} = C_{11} + C_m$$

(chap3.pdf, p.26)

- **Odd-mode characteristic impedance** (reduced):

$$Z_{\text{odd}} = \sqrt{\frac{L_{\text{odd}}}{C_{\text{odd}}}} = \sqrt{\frac{L_{11} - L_{12}}{C_{11} + C_m}}$$

(chap3.pdf, p.27)

- **Odd-mode time delay:**

$$\mathrm{TD}_{\mathrm{odd}} = \sqrt{L_{\mathrm{odd}} C_{\mathrm{odd}}} = \sqrt{(L_{11} - L_{12})(C_{11} + C_m)}$$

(chap3.pdf, p.27)
Effect: $Z_s = Z_0 > Z_{\mathrm{odd}}$ **underdriven** condition.

1.12 3.12 Even Mode Propagation *(pp.2829)*

When two adjacent lines carry **same-polarity signals** ($I_1 = I_2$, $V_1 = V_2$):

- **Even-mode inductance** (increased):

$$L_{\mathrm{even}} = L_{11} + L_{12}$$

(chap3.pdf, p.28)

- **Even-mode capacitance** (reduced):

$$C_{\mathrm{even}} = C_{11} - C_m$$

(chap3.pdf, p.29)

- **Even-mode characteristic impedance** (increased):

$$Z_{\mathrm{even}} = \sqrt{\frac{L_{11} + L_{12}}{C_{11} - C_m}}$$

(chap3.pdf, p.29)

- **Even-mode time delay:**

$$\mathrm{TD}_{\mathrm{even}} = \sqrt{(L_{11} + L_{12})(C_{11} - C_m)}$$

(chap3.pdf, p.29)
Effect: $Z_s = Z_0 < Z_{\mathrm{even}}$ **overdriven** condition.

1.13 3.13 Summary: Odd vs. Even Mode *(pp.3132)*

Property	Odd Mode	Even Mode
Signals	Opposite polarity	Same polarity
L_{eff}	$L_{11} - L_{12}$ (reduced)	$L_{11} + L_{12}$ (increased)
C_{eff}	$C_{11} + C_m$ (increased)	$C_{11} - C_m$ (reduced)
Z	Reduced ($< Z_0$)	Increased ($> Z_0$)
Driving condition	Underdriven	Overdriven
Time delay	Not obvious (TD may change either way)	Not obvious

Crosstalk does not induce velocity variations in a stripline, but **does** in a microstrip. *(p.32)*

1.14 3.14 Single-Line Equivalent Model (SLEM) *(pp.3335)*

For a multiconductor system, find the equivalent L and C seen by the target line (line 2), considering neighbors switching in phase or out of phase.

One signal out of phase with neighbors 1 and 3:

$$C_{\text{eq}} = C_{22} - C_{12} + C_{23}, \quad L_{\text{eq}} = L_{22} + L_{12} - L_{23}$$

(chap3.pdf, p.34)

Rules for SLEM: - Nearest neighbors have greatest effect (mutual parasitics fall off exponentially). - Common mode (all in phase) and differential mode (target out of phase) give worst-case impedance and velocity variations. - SLEM accuracy degrades when edge-to-edge spacing/height ratio < 1. *(p.35)*

1.15 3.15 Crosstalk Trends *(pp.3638)*

- **Lower-impedance** lines better EM field confinement less crosstalk-induced impedance variation.
- Mutual parasitics (L_m, C_m) fall off **exponentially** with trace-to-trace spacing.
- Even an undriven adjacent trace influences the impedance of a driven line.

1.16 3.16 Termination of Odd/Even Mode Line Pairs *(pp.3940)*

1.16.1 Pi-Termination Network:

- Even mode: $Z_{\text{even}} = 2R_3 \| R_1$ (parallel R_1 and two R_3 in series)
- Odd mode: $Z_{\text{odd}} = (1/2R_3) \| R_1$

1.16.2 T-Termination Network:

- Odd mode: $Z_{\text{odd}} = R_1$
- Even mode: $Z_{\text{even}} = R_1 + 2R_3$

(chap3.pdf, pp.3940)

1.17 3.17 Minimization of Crosstalk Rules of Thumb *(pp.4142)*

1. **Widen spacing** S between lines as much as routing allows.
2. **Minimize height** H above ground plane (tight coupling to ground less coupling to neighbors). While keeping target impedance.
3. Use **differential routing** for critical nets (e.g., system clock).
4. Route signals on different layers **orthogonally** to each other.
5. Use **stripline or embedded microstrip** to eliminate velocity variations.
6. **Minimize parallel run lengths** between coupled nets.
7. Place components to minimize trace congestion.
8. Use **slower edge rates** (with caution negative side effects on timing).